



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR

Symbiosis International (Deemed University)

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Experiment No. 9

Title: Implement localStorage and sessionStorage to save user preferences like theme settings.

Software/Tools Required: HTML5, CSS3, JavaScript, a web browser such as Google Chrome or Microsoft Edge, and a code editor such as Visual Studio Code.

Theory

1. Web Storage API

The Web Storage API provides client-side storage through localStorage and sessionStorage. It stores data as key-value pairs in the browser and allows JavaScript to save and retrieve information without sending it to a server.

2. LocalStorage

LocalStorage keeps data in the browser even after the page is closed or the browser is reopened. In this experiment, the dark-theme preference is stored in localStorage so that the selected dark theme can be retained for later visits.

3. SessionStorage

SessionStorage stores data only for the current browser tab or session. In the experiment, the white/light theme preference is stored using sessionStorage, demonstrating temporary preference storage during the active session.

4. Theme Preference

JavaScript reads the stored theme preference and applies the corresponding CSS class or styling to the page. If a saved preference exists, it is used when the page loads; otherwise, the default theme is applied. This makes the interface remember the user's selected appearance.

5. Page Redirection

After the preference is stored, the application redirects the user to the Student Login page. The login page reads the preferred theme from web storage and displays the interface using that theme, providing a consistent user experience.

Case Study Implemented

The supplied JavaScript case study demonstrates practical use of both storage mechanisms in a schedule planner. It saves schedule rows in localStorage under the key 'schedulePlannerRows', so inserted or deleted rows can be restored when the page is loaded. When a schedule row is selected, its day, time and topic are stored in sessionStorage under 'schedulePlannerSelectedRow', allowing the selected row to remain highlighted during the session. The script also recreates saved rows, binds Insert and Delete events, and provides a Reset action that removes both stored values and reloads the page.

Conclusion: The experiment successfully demonstrates persistent and temporary browser-side storage using localStorage and sessionStorage. The implemented theme preference is retained or used for the current session as required, and the Student Login page is opened with the preferred theme. The case study further demonstrates saving, retrieving, updating and clearing stored data in an interactive JavaScript application.